

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	05/08/2026	Duration:	60 minutes

Code of Conduct

I D.Yaswanth Sri Dana Ganesh (192425351) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Dana Ganesh

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
 - Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - Introduction
 - Problem Description

- Scope of the System
 - Functional Requirements
 - Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
5. **Identify Assumptions and Constraints**
- List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
- Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS)	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or

Documentation (15%)	organized, and includes all required sections.			lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

SOFTWARE REQUIREMENT SPECIFICATION (SRS)

Satellite Data Visualization Platform for Weather and Climate Pattern Analysis

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for the Satellite Data Visualization Platform, a web-based system that converts raw satellite imagery from public sources such as NASA and NOAA into accessible, interactive visualizations of weather and climate patterns. This document is intended for developers, project evaluators, and stakeholders to gain a common understanding of what the system must do.

1.2 Business Domain

The system operates in the meteorology and climate science domain, supporting weather forecasting, disaster preparedness, agricultural planning, and environmental research. It bridges the gap between complex scientific satellite data and non-expert decision-makers.

1.3 Document Conventions

Functional requirements are labelled FR1, FR2, ...; non-functional requirements are labelled NFR1, NFR2, ...; use cases are labelled UC1, UC2, ... Priority is indicated as High, Medium, or Low.

2. Problem Description

Satellite-derived weather and climate data is stored in complex scientific formats (e.g., GRIB, NetCDF, HDF5) that require specialised software and expertise to interpret. Meteorologists and researchers spend significant time manually processing raw imagery before patterns such as cyclone formation or temperature shifts can be identified. There is no intuitive, web-based platform that lets a broad range of users — from expert researchers to local planners and students — visualise historical and real-time satellite data through interactive maps and time-series charts, cross-reference multiple data layers, and receive timely alerts during extreme weather events. Existing tools tend to be expensive, proprietary, or too technically complex for smaller research teams, regional planning bodies, or students.

2.1 Stakeholders

Stakeholder	Interest / Role
Meteorologists / Researchers	Primary users who need fast pattern recognition and data export for scientific analysis.
Disaster Management Agencies	Need timely, processed visualizations and anomaly alerts to plan emergency response.
Government Planners / Policymakers	Use regional climate insights for agricultural, urban, and environmental planning.

Stakeholder	Interest / Role
Students / Academic Institutions	Use the platform for learning and research without needing specialist GIS tools.
System Administrators	Maintain data pipelines, manage users, and monitor system health.
Data Providers (NASA, NOAA)	External organisations supplying raw satellite imagery via public APIs.

2.2 Existing Challenges

- Satellite data requires specialised scientific software and technical expertise to interpret.
- Manual processing of raw imagery is time-consuming and delays pattern recognition.
- No integrated workflow exists for cross-referencing multiple data types (temperature, precipitation, cloud cover).
- Disaster agencies often lack timely access to processed visualizations during extreme weather events.
- Existing tools are expensive, proprietary, or too complex for students and small teams.

2.3 System Objectives

1. Develop a data ingestion pipeline to retrieve and process satellite imagery from NASA/NOAA.
2. Provide an interactive map-based visualization of real-time and historical weather patterns.
3. Implement time-series analysis tools for temperature, rainfall, and cloud cover.
4. Build a pattern-detection feature for anomalies such as cyclones, droughts, and temperature spikes.
5. Provide comparative visualization to overlay multiple data layers.
6. Enable data export for further research use.
7. Provide region-specific filtering.
8. Ensure the platform is usable through a simple web interface.

3. Scope of the System

The system will provide a web-based platform that ingests publicly available satellite data, processes it into analysis-ready form, and presents it through interactive maps, time-series charts, and comparative overlays. It will support region-based filtering, anomaly/pattern detection, data export, and role-based access for different stakeholder groups.

In Scope

- Ingestion of satellite data from NASA and NOAA public APIs.
- Interactive map visualization and time-series charting in a web browser.
- Automated anomaly/pattern detection (cyclones, droughts, temperature spikes).
- Layer overlay and comparison, region filtering, and data export (CSV/GeoTIFF/report).
- User authentication and role-based access (researcher, planner, administrator, public).

Out of Scope (Phase 1)

- Predictive/forecasting machine-learning models (identified as a future enhancement).
- Native mobile applications (initial release is web-only).
- Direct integration with private/commercial satellite data providers.

4. Functional Requirements

ID	Requirement	Priority
FR1	The system shall retrieve satellite imagery (infrared, visible, radar) from NASA/NOAA APIs on a scheduled and on-demand basis.	High
FR2	The system shall normalise raw scientific formats (GRIB, NetCDF, HDF5) into a common internal data model.	High
FR3	The system shall display an interactive map of real-time and historical weather patterns.	High
FR4	The system shall generate time-series charts of temperature, rainfall, and cloud cover for a selected period.	High
FR5	The system shall automatically detect anomalies such as cyclone formation, drought, or temperature spikes and raise alerts.	High
FR6	The system shall allow users to overlay and compare multiple data layers (e.g., precipitation vs. vegetation index).	Medium
FR7	The system shall allow authenticated users to export processed datasets and visualizations (CSV, image, GeoTIFF).	Medium
FR8	The system shall allow users to filter data by region (country, state, or custom bounding box).	High
FR9	The system shall provide role-based access control (researcher, planner, administrator, public user).	Medium
FR10	The system shall notify relevant users (e.g., disaster management officials) in real time when an anomaly is detected.	High
FR11	The system shall allow administrators to manage data source configurations and monitor pipeline health.	Medium
FR12	The system shall allow users to register, log in, and manage their account/profile.	Medium

5. Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	Map tiles and chart queries shall return results within 3 seconds for a standard regional query under normal load.
NFR2	Scalability	The system shall support increasing data volumes and concurrent users without architectural redesign, via horizontally scalable services.
NFR3	Reliability	The system shall recover gracefully from external API failures and retry failed ingestion jobs without data loss.
NFR4	Availability	The system shall maintain at least 99% uptime, with priority given to availability during active extreme-weather events.
NFR5	Security	All data in transit shall be encrypted (HTTPS/TLS); access to exports and admin functions shall require authentication and role-based authorisation.
NFR6	Usability	Non-expert users shall be able to generate a basic visualization within 3 clicks without prior GIS training.
NFR7	Maintainability	Individual modules (ingestion, analytics, visualization) shall be independently deployable and updatable.
NFR8	Interoperability	The system shall integrate with standard geospatial data formats (GeoJSON, GeoTIFF) and public REST APIs.
NFR9	Portability	The web interface shall function correctly on major modern browsers (Chrome, Firefox, Edge, Safari).

6. Actors and Use Cases

6.1 Primary Actors

- Meteorologist / Researcher — analyses weather/climate data in depth and exports datasets.
- Disaster Management Official — monitors alerts and regional conditions for emergency response planning.
- Regional Planner / Policymaker — uses regional visualizations for agricultural/urban planning.
- Student / General Public — browses visualizations for learning and general awareness.
- System Administrator — manages data pipelines, users, and system health.

6.2 Secondary Actors

- NASA / NOAA Data API — external system supplying raw satellite data.
- Alert Notification Service — external/internal service that dispatches real-time alerts (email/SMS/push).

6.3 Major Use Cases per Actor

Actor	Associated Use Cases
Meteorologist / Researcher	View Interactive Weather Map; Analyze Time-Series Climate Trends; Overlay & Compare Data Layers; Export Processed Data / Reports
Disaster Management Official	Detect Weather Anomalies / Pattern Alerts; Apply Region-Specific Filters; View Interactive Weather Map
Regional Planner / Policymaker	Apply Region-Specific Filters; View Interactive Weather Map; Analyze Time-Series Climate Trends
Student / General Public	View Interactive Weather Map; Analyze Time-Series Climate Trends; Login / Manage User Account
System Administrator	Manage Data Sources & Pipelines; Manage Users & Access Roles; Monitor System Health & Logs
NASA / NOAA Data API (secondary)	Ingest Satellite Data (Scheduled); Process & Normalize Raw Imagery
Alert Notification Service (secondary)	Send Real-Time Alert Notifications

6.4 Use Case Diagram

The diagram below shows the primary and secondary actors, the system boundary, and their associated use cases, including key <<include>> relationships.

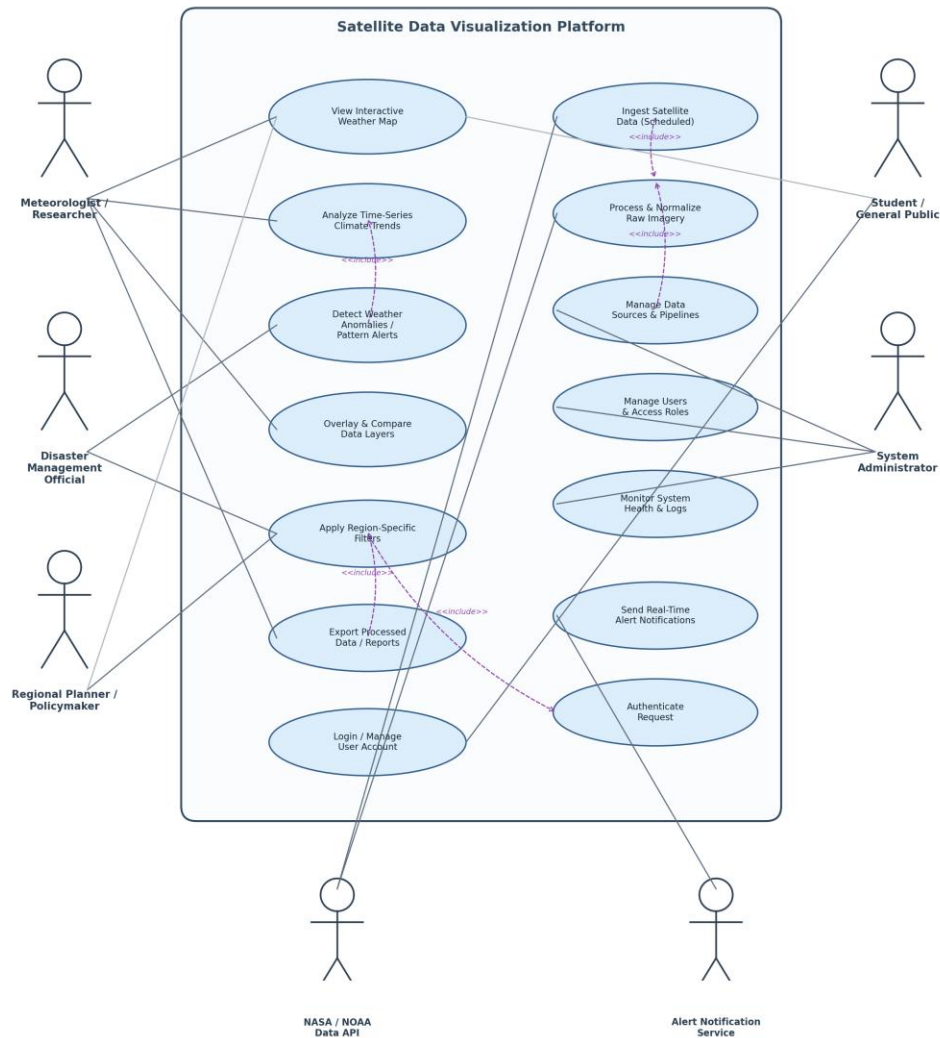


Figure 1: Use Case Diagram — Satellite Data Visualization Platform

7. Use Case Descriptions

Use Case ID	UC1
Name	View Interactive Weather Map
Actor(s)	Meteorologist, Disaster Official, Planner, Student
Description	User wants to see current or historical weather patterns on a map.
Precondition	User is logged in (or accessing public view) and selects a region and time.
Main Flow	1. User opens the map module. 2. User selects a region and date/time. 3. System retrieves processed data. 4. System renders the map with the selected imagery layer.
Postcondition	Map is displayed with the requested weather data overlay.

Use Case ID	UC2
Name	Analyze Time-Series Climate Trends
Actor(s)	Meteorologist, Planner, Student
Description	User wants to study how temperature, rainfall, or cloud cover changed over a selected period.
Precondition	Processed historical data is available for the selected region.
Main Flow	1. User selects a variable (temperature/rainfall/cloud cover) and a date range. 2. System queries the Processed Data Store. 3. System generates and displays a time-series chart.
Postcondition	Time-series chart is displayed and can be exported.

Use Case ID	UC3
Name	Detect Weather Anomalies / Pattern Alerts
Actor(s)	Disaster Management Official (system-triggered)
Description	System automatically flags unusual patterns that may indicate cyclones, droughts, or heat/cold spikes.
Precondition	Normalised satellite data is available in near real time.
Main Flow	1. System continuously analyses incoming data. 2. System applies anomaly-detection rules/models. 3. If an anomaly is found, system raises an alert and notifies subscribed users via the Alert Notification Service.
Postcondition	Relevant users receive a real-time alert with details of the detected anomaly.

Use Case ID	UC4
Name	Apply Region-Specific Filters
Actor(s)	Disaster Official, Planner
Description	User wants to narrow visualizations to a specific country, state, or custom area.
Precondition	User has access to the map or chart module.
Main Flow	1. User selects a region from a list or draws a custom boundary. 2. System applies the filter to subsequent map/chart/export requests.
Postcondition	Only data for the selected region is displayed.

Use Case ID	UC5
Name	Export Processed Data / Reports
Actor(s)	Meteorologist / Researcher
Description	User wants to download processed data or visualizations for offline analysis.
Precondition	User is authenticated and has selected a dataset/region/time range.
Main Flow	1. User selects export format (CSV, image, GeoTIFF). 2. System applies current filters. 3. System generates the export file. 4. System provides a download link.
Postcondition	Export file is downloaded successfully.

Use Case ID	UC6
Name	Manage Data Sources & Pipelines
Actor(s)	System Administrator
Description	Administrator wants to configure or troubleshoot data ingestion sources.
Precondition	Administrator is logged in with admin privileges.
Main Flow	1. Administrator views the list of configured data sources. 2. Administrator adds, edits, or disables a source. 3. System validates and applies the configuration.
Postcondition	Data source configuration is updated and ingestion continues accordingly.

8. Data Requirements

- Raw satellite imagery (infrared, visible, radar) with metadata: timestamp, satellite/sensor ID, geographic bounds, resolution.
- Normalised climate variables: temperature, precipitation, cloud cover, humidity, wind speed, linked to time and location.
- Derived/processed datasets: anomaly flags, trend summaries, comparative layer results.
- User data: account details, role, saved regions/preferences, export/download history.
- System/operational data: ingestion job logs, pipeline health metrics, alert history.

Data retention: raw imagery is retained in the data lake for reprocessing; processed/query-optimised data is retained in the time-series and spatial database for the platform's supported historical range (e.g., configurable, minimum 5 years for trend analysis).

9. External Interface Requirements

9.1 User Interfaces

- Responsive web application accessible via standard browsers, with a map view, chart view, filter panel, and export/alerts dashboard.

9.2 Hardware Interfaces

- No special hardware required for end users; server-side infrastructure requires sufficient compute/storage for imagery processing (cloud-based recommended).

9.3 Software Interfaces

- NASA Earthdata/EOSDIS API and NOAA satellite data API for data ingestion (REST/HTTPS).
- Mapping library (e.g., Leaflet/Mapbox) for interactive map rendering.
- Notification/email/SMS gateway for real-time alert delivery.

9.4 Communication Interfaces

- HTTPS for all client-server communication; WebSocket or similar for real-time alert push; internal service communication via REST/message broker.

10. System Features

10.1 Interactive Map Visualization

Displays real-time and historical satellite-derived weather data on a zoomable, pannable map with selectable imagery layers.

10.2 Time-Series Analytics

Generates trend charts for temperature, rainfall, and cloud cover over a user-selected date range and region.

10.3 Pattern/Anomaly Detection

Continuously scans incoming data for signs of cyclone formation, drought, or temperature anomalies, and raises real-time alerts.

10.4 Layer Comparison

Allows overlay of two or more data layers (e.g., precipitation and vegetation index) for comparative analysis.

10.5 Region Filtering & Export

Allows users to scope all visualizations to a specific region and export the resulting data/visualizations in common formats.

11. Assumptions and Constraints

11.1 Assumptions

- NASA and NOAA public APIs will remain available, free to access, and reasonably stable in structure.
- Users have a stable internet connection and a modern web browser.
- Satellite data volumes, while large, are within the processing capacity of a cloud-based infrastructure sized for this project.
- End users have basic digital literacy, even if they lack GIS/remote-sensing expertise.

11.2 Constraints

Type	Constraint
Technical	The system depends on the availability, rate limits, and data formats of third-party APIs (NASA/NOAA); changes to those APIs may require adapter updates.
Operational	Continuous, near-real-time ingestion requires reliable infrastructure monitoring and on-call support during extreme weather events.
Economic	Cloud storage and compute costs scale with imagery volume and user load; the project must operate within an allocated budget, which may limit historical data retention or processing frequency.
Legal	Use of NASA/NOAA data must comply with their respective open-data usage policies and attribution requirements.
Time	The project must be delivered within the academic/project timeline, which constrains the scope of the first release (see Section 3, Out of Scope).

12. Validation of Requirement Completeness and Consistency

12.1 Stakeholder Coverage Check

Each stakeholder group identified in Section 2.1 maps to at least one primary use case: researchers (UC1, UC2, UC5), disaster officials (UC3, UC4), planners (UC1, UC2, UC4), students/public (UC1, UC2), and administrators (UC6). No stakeholder identified in the scenario is left without a corresponding requirement, indicating good coverage.

12.2 Consistency and Ambiguity Review

- Checked for conflicting requirements: none found — e.g., NFR1 (3-second performance) and FR4 (time-series generation) are compatible if data is pre-aggregated rather than computed on demand.
- Checked for ambiguous terms: the original problem statement used vague terms like "quickly" and "timely"; these have been made measurable in NFR1 (3 seconds) and NFR4 (99% uptime).
- Checked for redundancy: FR3 (map visualization) and FR6 (layer overlay) were reviewed to confirm they are distinct capabilities rather than duplicates — FR3 is single-layer display, FR6 is multi-layer comparison.

12.3 Missing Requirements Identified During Review

- Added FR9 (role-based access control) and FR12 (account management), which were implied by "non-expert stakeholders" and multi-actor access but not explicitly stated in the original objectives.
- Added FR10 (real-time notification) to make the disaster-response objective actionable, since "timely access" implied a push mechanism rather than only on-demand viewing.
- Added NFR9 (portability/browser support) to make the "simple web interface" objective testable.

12.4 Suggested Improvements

- Define precise SLAs (e.g., exact uptime target, maximum acceptable data-ingestion lag) with stakeholders before development begins.
- Conduct usability testing with actual non-expert users (e.g., students, local planners) to validate NFR6.
- Clarify data retention and archival policy with legal/compliance input regarding NASA/NOAA data usage terms.
- Prioritise requirements with stakeholders using a formal technique (e.g., MoSCoW) to confirm the Priority column in Sections 4 and 5.

13. Conclusion

This SRS document defines a complete, validated set of functional and non-functional requirements, actors, use cases, and constraints for the Satellite Data Visualization Platform. It transforms the original problem statement into an actionable specification that addresses every identified stakeholder need while remaining consistent, measurable, and extensible for future enhancements such as predictive analytics.